Sime Darby Property University Hackathon 2026: Official Briefing Knowledge Base

> **Source Document:** `SDP University Hackathon Briefing with Student Nationwide.pdf` (37 Slides)
> **Briefing Date:** 9 October 2026 (6:30 PM via Microsoft Teams)
> **Organizers:** Sime Darby Property Berhad in partnership with Kolaxus Innovation Platform
> **Compiled For:** AI Agents & Engineering Team quick-reference guide

1. Executive Summary & Core Hackathon Profile

Parameter	Official Specification
Hackathon Title	Sime Darby Property University Hackathon 2026: *Township of Tomorrow*
Theme / Core Question	*"How might we reimagine the Township of Tomorrow where people, businesses, and nature thrive together through innovation, technology, and sustainable living?"*
Primary Tracks	**1. PEOPLE** (Healthier, safer, connected, fulfilling living) **2. PLANET** (Environmental regeneration, effortless sustainable living) *(Teams must select ONE focus area)*
Team Composition	3 to 5 university students (multidisciplinary skills encouraged)
Total Prize Pool	**RM 14,000** cash + Priority Internship & Management Associate Programme (MAP) placement
Key Partners & Hosts	• **Dato' Seri Azmir Merican** (Group Managing Director & CEO, SDP) • **Dr Lam Su Ki** (Head, Innovation & Partnership, SDP) • **Elmen Tan** (Chief Digital & Technology Officer, SDP) • **Shahidah Salam (Dina)** (Manager, Innovation & Partnership, SDP) • **Mr Sivavenayakam** (Managing Director, Kolaxus)

2. Critical Milestones & Timeline (Strict Deadlines)

> [!IMPORTANT]
> **Physical attendance is mandatory** for all selected finalists for both the Top 10 Pitch (14 Nov) and the Final Award Ceremony (3 Dec). Students must arrange and bear their own travel and accommodation costs.

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flowchart LR
    A["9 OCT<br>Briefing"] --> B["21 OCT<br>Registration Closes"]
    B --> C["23 OCT<br>Proposal Submission (11:59 PM)"]
    C --> D["24–28 OCT<br>Judging & Evaluation"]
    D --> E["28 OCT<br>Top 10 Announced"]
    E --> F["2–6 NOV<br>Mentoring Sprint"]
    F --> G["11 NOV<br>Final Submission"]
    G --> H["14 NOV<br>Physical Pitch & Talent Spotlight"]
    H --> I["17 NOV<br>Top 5 Notified (Rankings Hidden)"]
    I --> J["3 DEC<br>Final Presentation & Award Ceremony (Elmina)"]
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Date	Milestone	Mandatory Details & Requirements
9 Oct 2026	**Challenge Briefing**	Virtual launch on Microsoft Teams (6:30 PM).
21 Oct 2026	**Registration Closes**	**CRITICAL DEADLINE 1:** All team members (3–5 pax) must be registered.
23 Oct 2026	**Proposal Submission**	**CRITICAL DEADLINE 2 (11:59 PM):** Incomplete submissions will NOT be considered.
24–28 Oct 2026	**Evaluation & Judging**	Expert screening of all submitted proposals against 6-axis rubric.
28 Oct 2026	**Top 10 Announcement**	Top 10 selected finalist teams announced.

Date	Milestone	Mandatory Details & Requirements
2–6 Nov 2026	**Top 10 Mentoring Sprint**	Dedicated sessions with Industry Subject Matter Experts to refine prototype, pitch, and business case.
11 Nov 2026	**Final Submission**	Refined pitch deck and demo assets submitted prior to physical pitch.
14 Nov 2026	**Pitch to Judges & Talent Spotlight**	**Compulsory physical attendance** at Sime Darby Property Office (9:30 AM – 5:00 PM). • **Morning (9:30 AM – 1:00 PM):** 10-min Pitch + 5-min Q&A per team. • **Afternoon (2:30 PM – 5:30 PM):** Individual 5–10 min Talent Spotlight interviews.
17 Nov 2026	**Top 5 Notified**	Top 5 selected teams notified. *Final rankings remain hidden until 3 Dec.*
3 Dec 2026	**Final Presentation & Award Ceremony**	**Compulsory physical attendance** at Elmina, Sime Darby Property. • **Top 5:** Pitch to leadership audience. • **Top 10:** Dedicated showcase booth for live demonstration. • Winners announced and trophies/cash awarded.

3. Four Mandatory Submission Components (Due 23 Oct, 11:59 PM)

All teams must submit through the official portal. An incomplete package leads to immediate disqualification:

#	Component	Requirement Level	Specifications & Constraints
1	**Pitch Deck**	**REQUIRED**	**Maximum 10 main slides** covering the 8 required core areas. Appendix slides are permitted for secondary/technical details. Format: PDF.
2	**Video / Demo Link**	**REQUIRED**	**5-minute video** or live demo link showcasing and proving the working solution.
3	**Prototype / Code**	**REQUIRED**	Link to working prototype, codebase, or GitHub repository.
4	**Supporting Documents**	*OPTIONAL*	Additional research, technical documentation, validation data, or letters of support.

4. Pitch Deck Architecture (The Required 8-Slide Structure)

The briefing mandates an 8-topic flow divided into **The Story** and **The Proof**, capped at **10 core slides**:

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│ THE STORY (Slides 1–4)                                  │
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│ 1. Problem       │ Clear, real township friction point  │
│ 2. Target User   │ Specific resident or operator persona│
│ 3. Solution      │ Proposed intervention / product      │
│ 4. Innovation    │ Differentiation from existing tools  │
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│ THE PROOF (Slides 5–8)                                  │
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│ 5. How It Works  │ Architecture, data flow & tech stack │
│ 6. Impact        │ Measurable resident/planet outcomes  │
│ 7. Feasibility   │ Deployment ease, economics, safety   │
│ 8. Roadmap       │ Phased pilot plan & SDP scaling path │
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* Appendix: Technical schematics, sensor physics, math models, raw data charts
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5. Official Judges Evaluation Rubric (6 Axes / 100%)

Judges evaluate every submission against 6 specific criteria with distinct weightages:

Axis	Weight	Primary Guiding Question	What Judges Look For
1. Impact & Sustainability Value	**25%**	*Can the solution create meaningful and measurable resident, community or environmental outcomes?*	Quantified metrics (kWh saved, carbon avoided, cost reduction, waste diverted, hours saved).
2. Innovation & Differentiation	**20%**	*Is there genuine originality or a differentiated approach rather than simply repackaging an existing solution?*	Novel technology combinations, edge computing, multimodal data fusion vs off-the-shelf single sensors.
3. Feasibility & Deliverability	**20%**	*Could the solution realistically be piloted and implemented within a township environment?*	Low BOM cost, non-invasive installation, robust hardware/firmware, no disruptive structural retrofits.
4. Strategic Fit & Theme Relevance	**15%**	*Does the idea clearly address a People or Planet challenge and an SDP township priority?*	Direct applicability to SDP developments (Elmina, KL East, KLGCC, Bukit Jelutong, Bandar Bukit Raja).
5. Commercial & Value Multiplication	**10%**	*Could the solution create value for SDP, township businesses and/or residents and potentially scale?*	Viable business model (B2B FM savings + B2C subscription), clear ROI, value multiplier concept.
6. Team, Evidence & Pitch Quality	**10%**	*Has the team researched and validated the problem, developed a credible solution and communicated it clearly?*	Working prototype, empirical real-world datasets, clear articulation, high credibility.

6. Challenge Themes & Sub-Track Breakdown

Teams must select **ONE primary theme** (People OR Planet):

Theme A: PEOPLE

> *"How might we enable every resident to live healthier, safer, more connected and more fulfilling lives?"*
> - **01. Community Wellbeing:** Physical and mental health of township residents.
> - **02. Safety & Security:** Safer streets, smart surveillance, physical security.
> - **03. Accessibility & Inclusion:** Universal design for elderly, disabled, and children.
> - **04. Mobility:** Micro-mobility, pedestrian pathways, seamless transit connectivity.
> - **05. Digital Living:** Smart home integration, predictive utilities, friction-free services.
> - **06. Neighbourhood Connection:** Inter-neighbour social capital and community bonding.
> - **07. Healthy Lifestyles:** Active living, parks utilization, sports engagement.

Theme B: PLANET

> *"How might we create townships that regenerate the environment while making sustainable living effortless for communities, residents and businesses?"*
> - **01. Waste Management:** Smart sorting, collection routing, organics reduction.
> - **02. Circular Economy:** Closed-loop resource recycling, repair/reuse systems, asset longevity.
> - **03. Energy Efficiency:** Parasitic load elimination, HVAC optimization, renewable micro-grids.
> - **04. Water Conservation:** Leak detection, rainwater harvesting, greywater recycling.
> - **05. Green Transport:** EV charging infrastructure, low-carbon township transit.
> - **06. Biodiversity:** Flora/fauna conservation, urban nature corridors, reforestation.
> - **07. Climate Resilience:** Flood mitigation, urban heat island cooling, storm readiness.

7. Prize Structure & Career Acceleration Pathway

Cash Prizes (RM 14,000 Total)

- 🥇 **Champion: RM 6,000**
- 🥈 **1st Runner-Up: RM 3,000**
- 🥉 **2nd Runner-Up: RM 1,500**
- 🏅 **Consolation (4th to 10th Place): RM 500 each** (7 teams = RM 3,500)

Beyond Cash: The Talent Pipeline

- **Internship Placement:** Up to 10 selected top performers receive internship offers at Sime Darby Property to continue developing their project alongside SDP mentors.
- **Management Associate Programme (MAP):** Fast-track access to SDP's flagship leadership development track for standout graduates.
- **PoC Incubation:** Selected solutions enter SDP's innovation funnel for live township piloting.

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8. Afternoon Talent Spotlight (14 Nov 2026)

On the Top 10 Pitch Day, each student undertakes an individual **5–10 minute interview** with SDP HR and leadership. The briefing outlines the exact questions students must prepare:

1. **Who are you beyond your resume?**
2. **What is the one trait that defines you the most?**
3. **What makes you different from others?**
4. **What is your genuine passion?**
5. **What specific role and contribution did you deliver to your team's solution?**
6. **What achievement or experience in your life are you most proud of?**

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9. Legal Terms, PDPA & Intellectual Property (IP)

> [!CAUTION]
> **Key Intellectual Property Clause (Slide 32):**
> *"All submissions and solutions created during the hackathon become the intellectual property of the Organiser (Sime Darby Property). The Organiser has full rights to use, reproduce, modify, showcase and commercialise all submitted materials."*

- **Plagiarism Policy:** Strictly prohibited. Proper citations required for any third-party code, open-source models, or external datasets.
- **PDPA / Privacy:** Participants consent to personal data collection for event operations, background checks, and talent recruiting.
- **Media Rights:** SDP reserves rights to photograph, record, and broadcast all team presentations and prototype footage.

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10. Strategic Mapping: How Our Project Fits the Hackathon

> Numbers here follow `docs/proposals/PROPOSAL_WRITING_GUIDE.md` §4 (single source, synced 2026-10-20). Unmeasured impact figures are deliberately absent; see that guide's Say / Plan / Don't-say table.

Our project (**Multimodal IoT Predictive Maintenance Platform**) aligns with the briefing requirements through the following positioning:

Hackathon Requirement	Our Implementation & Evidence
Primary Theme Selection	**PLANET $\rightarrow$ Track 02: Circular Economy & Track 03: Energy Efficiency** *(Supported by People $\rightarrow$ Track 05: Digital Living)*
The "Micro to Macro" Story	The domestic washing machine serves as an aggressive **high-stress micro testbed** (severe unbalance, transient water dynamics, rapid acceleration). Validating multimodal sensor fusion here is the first step toward **macro township infrastructure** (central HVAC compressors, chillers, and water supply booster pumps in mosques, clubhouses, and Elmina Lakeside Mall).
Impact (25%)	Circular-economy case: catching wear early lets a small part be replaced instead of the whole machine (illustrative scenario, not yet measured; pilot to confirm). Energy case is a hypothesis: one labelled estimate only (5 kW AC, 10% extra draw, about RM 900/yr). No measured ringgit, kWh, carbon or scrap figures yet.

Hackathon Requirement	Our Implementation & Evidence
Innovation (20%)	Early feature fusion of vibration (MPU-6050 @ 200 Hz), sound (INMP441 @ 16 kHz, on-device DSP so no recordings are uploaded) and power (Tapo P110 @ 1 Hz) with a per-machine one-class baseline that needs no fault labels. SmartThings phase context is supported in code but not yet active in the evaluated model.
Feasibility (20%)	Non-invasive external mount (3M VHB tape today; a magnetic mount is a proposal). Pilot kit RM 169.68 net paid (node RM 131.07 + RM 38.61 smart plug; see `HARDWARE_BOM_AND_COST_BREAKDOWN.md`); sub-RM 50 is an unvalidated volume target. FreeRTOS dual-core firmware 07 is running.
Deliverables Proof	• **Codebase:** Fully documented repo with modular firmware, Python streaming pipelines, and ML models. • **Empirical Data:** 11 recorded wash sessions (8,379 standardized rows, as of 20 Oct), including 2 complete firmware-07 cycles; 8 healthy earlier-firmware cycles in the nested leave-one-cycle-out evaluation: 0 of 8 falsely flagged (95% upper bound 31%), simulated sound fault caught in 7 of 8. • **Live Demo:** Web dashboard deployed at `washer-pdm.pages.dev`.

11. Quick Contacts & Support Channel

- **Organizing Partner:** Kolaxus Innovation Platform (`support@kolaxus.net`)
- **SDP Innovation Team:** Dr Lam Su Ki, Shahidah Salam (Dina)
- **Official Communication:** Dedicated hackathon support channel for FAQs, clarifications, and submission checks.